

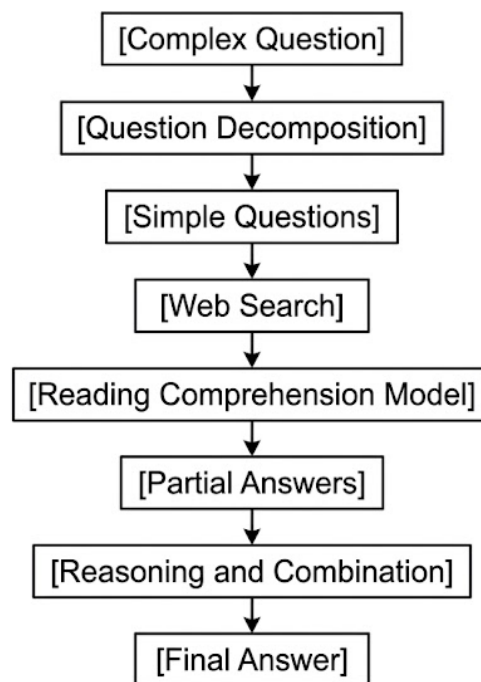
WebQuestionsSP (ACL 2016)

It introduces a new method for answering complex questions by dividing them into smaller and simpler questions. Instead of solving the entire question at once, the system answers each simple question separately and combines the results to produce the final answer.

This approach is called **Question Decomposition**, and it became an important idea in the development of multi-hop question answering systems.

Proposed Method

The authors solve this problem by dividing a complex question into multiple simple questions.



Example

Original Question:

What city is the birthplace of the author of "Without End" and hosted Euro 2012?

The system breaks it into smaller questions:

1. Who is the author of "Without End"?
2. Where was the author born?
3. Which cities hosted Euro 2012?

After collecting all the answers, the system compares them and finds the common city.

Final Answer:

Lviv

Types of Reasoning

The paper supports four major reasoning operations:

1. Composition

The answer from one question becomes the input for another question.

Example:

Who wrote Harry Potter?

↓

J.K. Rowling

↓

Where was J.K. Rowling born?

2. Conjunction

The system finds the common answer between two conditions.

Example:

Movies starring Tom Cruise

AND

Directed by Steven Spielberg

3. Superlative

Questions involving the highest, largest, oldest, etc.

Example:

Which is the highest mountain in Europe?

4. Comparative

Questions comparing two values.

Example:

Which countries are larger than Germany?

Results

The proposed method achieved better performance than directly answering the entire question.

Method	Precision@1
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Simple QA	20.8%
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SplitQA	27.5%
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Human Performance 63.0%

Limitations

- Depends on Google Search results.
- Supports only one level of question decomposition. Thank you for your request. Please provide the text you would like rewritten in a professional manner so I can accurately revise it for tone, clarity, and style. Include any specific preferences such as desired level of formality, audience (e.g., executive, client, internal team), word limit, or any key points that must remain unchanged. Once I have the original text and those preferences, I will deliver a polished, professional version along with brief notes explaining the changes made.
- Does not handle deep reasoning very well.
- Training uses automatically generated labels, which may contain errors.

Conclusion

This paper introduced the idea of **Question Decomposition** for answering complex questions. Instead of solving a difficult question directly, it divides the question into simpler sub-questions, retrieves answers from the web, and combines them to produce the final result. Experimental results show that this method performs better than traditional QA systems and has influenced later multi-hop question answering datasets such as **HotpotQA** and **2WikiMultihopQA**.

WebQuestionsSP (ACL 2016)

This paper presents a method for answering complex questions by decomposing them into smaller, simpler sub-questions. Rather than attempting to resolve a multi-faceted query in a single step, the approach solves each sub-question independently and combines their answers to generate the final response. This strategy, referred to as Question Decomposition, has become a foundational concept for modern multi-hop question answering systems.

Proposed Method

The authors address complex question answering by splitting an input question into multiple simpler questions. Each sub-question is answered separately, typically via web retrieval, and the partial answers are composed to obtain the overall solution.

Example

Original question: What city is the birthplace of the author of "Without End" and hosted Euro 2012?

Decomposition:

1. Who is the author of "Without End"?
2. Where was that author born?
3. Which cities hosted Euro 2012?

After gathering the answers for each sub-question, the system identifies the intersection and returns the city Lviv as the final answer.

Types of Reasoning Supported

The paper formalizes four primary reasoning operations that the decomposition method supports:

1. Composition

The output of one sub-question is used as the input to another. Example: "Who wrote Harry Potter?" → "J.K. Rowling" → "Where was J.K. Rowling born?"

2. Conjunction

Answers satisfying multiple conditions are identified by intersecting result sets. Example: Films starring Tom Cruise AND directed by Steven Spielberg.

3. Superlative

Questions that require selecting extreme values, such as highest, largest, or oldest. Example: "Which is the highest mountain in Europe?"

4. Comparative

Questions that compare entities on a measurable attribute. Example: "Which countries are larger than Germany?"

Results

Experimental evaluation shows decomposition yields improved performance compared with attempting to answer complex questions directly. Reported Precision@1 scores: Simple QA 20.8%, SplitQA 27.5%, Human performance 63.0%.

Limitations

- Reliance on web search results (Google) for retrieval.
- Supports only a single level of decomposition in the presented system.
- Limited capacity for deep, iterative reasoning.
- Training labels are automatically generated and may include errors.

Conclusion

The paper introduces Question Decomposition as an effective strategy for multi-hop question answering. By splitting complex queries into tractable sub-questions, retrieving answers, and composing results, the method outperforms straightforward QA baselines and influenced subsequent datasets and systems such as HotpotQA and 2WikiMultihopQA.

Source: [Click Here](#)

Talmor, A., & Berant, J. (2018, June). The web as a knowledge-base for answering complex questions. In Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers) (pp. 641-651).